

ITEX 120 Probability and Statistical Modeling

Course Title: Probability and Statistical Modeling

Course Code: ITEX 120

Credit Hours: 3 (2 TH + 1 PR)

Nature: Theory + Practical

Course Description and Goals

This course aims to explore the application of probability and statistics for computer application. Students will use the techniques and strategies of statistical modeling and apply in the computing field. The objective of the course is to provide students with a clear understanding of the basic statistical concepts and tools and to enable them to use these tools as Necessary Avenue for engineering professions and scientific knowledge.

Learning Outcomes

On completion of this course, the students will be able to:

- Apply probability principles to model uncertainty in IT systems.
- Analyze discrete and continuous random variables and compute expectations.
- Use probability distributions to model real-world data.
- Summarize and visualize datasets using statistical techniques.
- Apply correlation and regression for predictive modeling.
- Perform sampling and estimation for population parameters.
- Conduct hypothesis testing for data-driven decision making.
- Apply statistical quality control in system monitoring and optimization.
- Use computational tools (Python, Excel, SPSS, R) for statistical analysis.
- Interpret statistical results in real-world IT and data science problems.

Course Contents

Specific Objectives	Contents
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• Understand fundamental concepts of probability and uncertainty. • Apply set theory operations to probability problems. • Define and construct sample spaces and events. • Differentiate between mutually exclusive, independent, and exhaustive events. • Apply probability axioms and addition/multiplication rules. • Compute conditional probabilities. • Apply the Law of Total Probability. • Use Bayes' Theorem for decision-making and classification problems. • Model simple IT-related risk and decision scenarios using probability.	Unit 1: Probability (6 hours) • Introduction • A review of sets • Basic terminology in probability • Laws of Probability • Conditional probability • Partitions, total probability • Bayes' theorem and its applications
• Define discrete and continuous random variables. • Construct probability mass and density functions. • Compute cumulative distribution functions (CDF). • Calculate mean, variance, and standard deviation. • Interpret mathematical expectation in practical contexts. • Apply expectation to model average system performance and outcomes.	Unit 2 : One Dimensional Random Variables and Mathematical Expectation (5 hours) • Introduction • The Distribution Function • Discrete and Continuous Random variable and its function • Some characteristics of distributions (mean and variance) • Equivalent events • Mathematical expectation.
• Understand Bernoulli trials and Bernoulli distribution. • Apply the Binomial distribution to discrete event modeling. • Compute mean and variance of Binomial distribution. • Apply Poisson distribution. • Approximate Binomial distribution using Poisson distribution. • Understand properties of the Normal	Unit 3 : Probability Distribution (8 hours) • Introduction • Bernoulli Trials and Bernoulli Distribution • Binomial Distribution - Mean and variance - The cumulative binomial distribution - Application of binomial distribution

distribution. • Standardize variables using Z-scores. • Use Normal approximation to Binomial distribution. • Model real-world IT systems using appropriate distributions.	- Fitting of a binomial distribution • Poisson distribution - Mean and variance - The cumulative poisson distribution - Application of poisson distribution - The poisson approximation to binomial distribution - Fitting of poisson distribution • Normal Distribution - Properties of the normal distribution - Mean and variance - Standard Normal Distribution - Problem-solving Procedure - The normal approximation to binomial distribution
• Interpret the importance of statistics in IT and data science. • Organize and summarize raw data effectively. • Construct and interpret graphical representations of data. • Compute measures of central tendency (mean, median, mode, partition). • Calculate measures of dispersion (range, variance, standard deviation). • Interpret coefficient of variation. • Analyze skewness and kurtosis of datasets. • Compare datasets using numerical and graphical methods.	Unit 4 : Introduction to Statistics and Data Description (8 hours) • Graphical Presentation of Data - Dot plots and scatter plots - Frequency distribution and histogram - The stem and leaf plot - The box plot - The pareto chat • Numerical Description of Data - Measures of Central Tendency - Measures of Dispersion - Skewness and Kurtosis
• Understand correlation and its interpretation. • Calculate and interpret the correlation coefficient. • Develop simple linear regression models. • Estimate regression coefficients using least squares method. • Interpret slope and intercept in real-world context.	Unit 5 : Simple Linear correlation and regression (5 hours) • Correlation and interpretation • Probable error and significance • Coefficient of determination • Simple linear regression and interpretation

• Compute and interpret coefficient of determination (R^2). • Use regression for prediction in IT and data analysis problems.	
• Distinguish between population and sample. • Understand the concept of parameters and statistics. • Describe properties of sampling distributions. • Apply the Central Limit Theorem in sampling contexts. • Apply t, Chi-square, and F distributions. • Perform point estimation of parameters. • Construct confidence intervals for mean and variance. • Construct two-sample confidence intervals. • Evaluate properties of estimators (bias, efficiency, consistency). • Apply estimation techniques in performance analysis and system evaluation.	Unit 6 : sampling and Estimation (5 hours) • Sampling techniques • Sampling and non-sampling errors • Sampling distribution • Standard error and application • Central limit theorem • Estimation theory • Properties of estimators • Point and interval estimation • Single and two sample confidence interval estimation (mean and variance)
• Formulate null and alternative hypotheses. • Understand Type I and Type II errors. • Conduct hypothesis testing for single population mean and variance. • Conduct hypothesis testing for two population means and variances. • Interpret p-values and decision rules. • Apply hypothesis testing in IT.	Unit 7: Tests of Hypothesis (5 hours) • Introduction • Types of hypothesis (null and alternative) • Errors (Type I and Type II) • Level of significance, critical value and region, • Tests hypothesis on single sample (mean and variance) • Tests of hypothesis on two samples (mean and variance)
• Understand the principles of statistical process control.	Unit 8: Statistical Quality Control (3 hrs) • Introduction, Statistical Process control

• Construct and interpret control charts for measurements. • Construct control charts for individual observations. • Construct control charts for attributes. • Detects process variation and abnormal behavior. • Apply statistical quality control in IT system monitoring and reliability analysis.	• Control charts for measurements • Control charts for individual measurements • Control chart for attributes
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Theory Sessions

The theory component of the course will be delivered using a combination of interactive, analytical, and technology-supported teaching strategies to promote conceptual understanding and practical application.

1. Concept-Based Lectures

- Structured lectures explaining probability and statistical concepts with formal definitions and derivations.
- Emphasis on conceptual clarity before procedural techniques.
- Step-by-step explanation of formulas and theorems.

2. Interactive Problem-Solving Sessions

- Guided problem-solving demonstrations.
- Real-time worked examples involving probability models and statistical inference.
- Classroom exercises with incremental difficulty levels.

3. Application-Oriented Discussions

- Case studies linking probability and statistics to IT domains such as data science, machine learning, system reliability, and quality control.
- Discussion of real-world datasets and decision-making scenarios.

4. Collaborative Learning

- Think–Pair–Share activities for conceptual reinforcement.
- Small group discussions on statistical interpretation and hypothesis testing.

- Peer explanation of statistical reasoning.

5. Technology-Integrated Teaching

- Demonstrations using Python (NumPy, SciPy, Pandas, Matplotlib), Excel, or SPSS.
- Visualization of probability distributions and statistical models.
- Simulation of random experiments and sampling processes.

6. Inquiry-Based Learning

- Students explore probability experiments through guided discovery.
- Analytical questioning to develop statistical reasoning.
- Encouragement of hypothesis formulation and testing.

7. Concept Reinforcement Techniques

- Short quizzes and formative assessments.
- Summary sessions at the end of each unit.
- Error analysis of common mistakes in probability and inference.

8. Mathematical Communication Development

- Students explain solutions verbally and in written form.
- Interpretation-focused discussions rather than formula memorization.
- Emphasis on reasoning, justification, and clarity in statistical conclusions.

Computational Labs (Minimum 15 Hours)

Lab 1: Probability Simulation (2 hours)

- Simulate random experiments using Python.
- Compute empirical probabilities.
- Verify theoretical probability results using simulation.
- Apply Monte Carlo simulation techniques.

Tools: Python (NumPy, random module)

Lab 2: Random Variables and Distributions (2 hours)

- Generate discrete and continuous random variables.
- Visualize PMF and PDF.
- Calculate mean and variance computationally.
- Compare theoretical and simulated results.

Tools: Python (NumPy, SciPy, Matplotlib)

Lab 3: Data Visualization and Descriptive Statistics (3 hours)

- Import and clean real datasets.
- Construct histograms, box plots, scatter plots.
- Compute mean, median, standard deviation, coefficient of variation.
- Interpret graphical results.

Tools: Python (Pandas, Matplotlib) / Excel / SPSS

Lab 4: Probability Distributions Modeling (2 hours)

- Model Binomial and Poisson distributions.
- Normal distribution visualization and Z-score computation.
- Apply Central Limit Theorem through simulation.

Tools: Python / SPSS

Lab 5: Regression and Correlation Analysis (2 hours)

- Compute correlation coefficient.
- Fit simple linear regression model.
- Interpret regression parameters.
- Visualize a regression line.

Tools: Python / SPSS

Lab 6: Estimation and Hypothesis Testing (2 hours)

- Construct confidence intervals.
- Perform one-sample and two-sample tests.
- Interpret p-values and statistical decisions.

Tools: Python / SPSS

Lab 7: Statistical Quality Control (2 hours)

- Construct control charts (mean, range, attribute charts).

- Detects process variation.
- Analyze real or simulated process data.

Deliverables

- Lab reports (for each lab)
- Programming scripts or statistical analysis files
- Final project report (5–10 pages)
- 10–15 minute presentation
- Demonstration of analysis

Assessment Plan

• In-semester (50)

Component	Marks	Description
Assignments	15	Weekly problem sets (8-10 assignments)
Quizzes	10	Short quizzes (4-6 quizzes, lowest dropped)
Lab Work & Reports	15	Practical lab activities and documentation
Project & Presentation	10	Final IT application project
Total	50	

• End-semester (50)

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (MCQ): 10 Questions × 1 marks
- Section B (Short Answer Question-SAQ) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (Long Answer Question-LAQ) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SAQs	LAQs
Unit One: Probability	6	1	1	
Unit Two: One Dimensional Random Variables and Mathematical Expectation	5	1	1	
Unit Three: Probability Distribution	8	2	1	1

Unit Four: Introduction to Statistics and Data Description	8	2	1	1
Unit Five: Simple Linear correlation and regression	5	1	1	-
Unit Six: Sampling and Estimation	5	1	1	1
Unit Seven: Tests of Hypothesis	5	1		1
Unit eight: Statistical Quality Control	3	1	1	
Total	45	10	7	3

References:

Hines, W. W., Montgomery D. C., Goldsman, D. M., Borror, C. M. (2003). *Probability and statistics in engineering* (4th ed.). John Wiley and Sons, Inc.